SKKU 2024 Challenge: Heisenberg Spin Glass

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Part I : A first look at the Heisenberg spin glass

How many symmetry classes are there?

There are 11 distinct symmetry classes, each characterized by unique eigen-spectra. I conducted eigen-decomposition for all 64 possible combinations of the coefficients J_{ij}.

The following code generates the Heisenberg Hamiltonian $H_{heisenberg}$:

```
def __reset__(self):
    qubit_idx = [idx for idx in range(self.num_qubits)]
    combinations = itertools.combinations(qubit_idx, 2)

    H_z = np.zeros([2**self.num_qubits, 2**self.num_qubits]
                   , dtype=np.complex128)
    if self.coefficients2 is not None:
        for i in range(self.num_qubits):
            temp = [self.I for _ in range(self.num_qubits)]
            temp[i] = self.Pauli_Z
            H_z += self.coefficients2[i]*self._Kron(temp)

    H_heisenberg = np.zeros([2**self.num_qubits,
                            2**self.num_qubits]
                            , dtype=np.complex128)
    for coefficient, pair_idx
      in zip(self.coefficients1, combinations):
        H_heisenberg += coefficient * self._H_ij(pair_idx)
    self.H_sg = H_heisenberg + H_z
```

In this example, `self.coefficients2` is set to `None`. Within the `for` loop, each pair of qubit indices (i, j) runs through unique combinations in the following sequence: $(0, 0)$, $(0, 1)$, $(0, 2)$, $(1, 2)$, $(1, 3)$, $(2, 3)$. This setup ensures that each distinct interaction between qubits is included without repetition, covering all possible unique two-qubit interactions in the Heisenberg

Hamiltonian $H_{heisenberg}$. Additionally, the `_Kron` method is implemented to maintain the Qiskit bit-ordering convention.

```
def _Kron(self, array):
    prod = np.array(1)
    if len(array) == 0:
        return prod
    else:
        for element in array:
            prod = np.kron(element, prod)
        return prod
```

Two models are classified within the same symmetry class if their eigenspectra are identical. Here, the eigenspectrum of a Hamiltonian represents the set of its eigenvalues. To numerically determine whether two eigenspectra are equivalent, I used the `np.allclose` function.

```
symmetry_classes = {}
for coefficient_combination
  in itertools.product([1, -1], repeat=6):

    eigenspectrum = H_sg(coefficient_combination, [0, 0, 0, 0])
    symmetry_classes[coefficient_combination] = eigenspectrum

classified_groups = {}
unique_eigenspectra = {}
class_count = 0

for combination, spectrum in symmetry_classes.items():
  is_duplicate = False
  for class_id, class_spectrum in unique_eigenspectra.items():
    if np.allclose(class_spectrum, spectrum._eigenSpectrum()):
      is_duplicate = True
      classified_groups[class_id].append(combination)
      break

  if is_duplicate:
    continue
  else:
    class_count += 1
    classified_groups[class_count] =[combination]
    unique_eigenspectra[class_count]=spectrum._eigenSpectrum()
```

Detailed information on eigenspectra and coefficient values can be found in the next questions.

For each symmetry class, provide one set of J_{ij} coefficients for a model in that symmetry class.

The Heisenberg Hamiltonian $\mathcal{H}_{\text{heisenberg}}$ is defined as:

$$\mathcal{H}_{\text{heisenberg}} = \sum_{(i,j)} J_{i,j} H_{i,j} \tag{1}$$

where

$$H_{i,j} = \sigma_x^i \sigma_x^j + \sigma_y^i \sigma_y^j + \sigma_z^i \sigma_z^j \tag{2}$$

The interaction is ferromagnetic when $J_{ij} = 1$ and anti-ferromagnetic when $J_{ij} = -1$. This Hamiltonian pocesses intrinsic symmetry: Interchange symmetry. Under the interchange interaction on qubit i and j, both of Hamiltonian eigenvalues and eigenstates remain unchanged. Therefore, the elements in the same symmetry class can be generated via interchange operation.

The blue edge denotes ferromagnetic interaction and red edge denotes anti-ferromagnetic interaction.

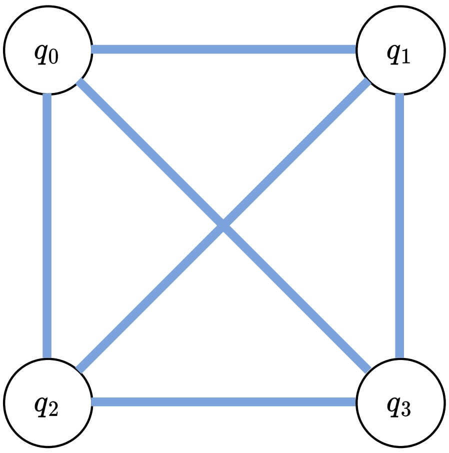

Figure 1: caption

Class 1. The whole interactions are ferromagnetic. There is no other element even by using interchange interaction.

Class 2

Class 3

Class 4

Class 5

Class 6

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	1	1	1	1	1

Table 1: Example LaTeX Table

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	1	1	1	1	-1
1	1	1	1	-1	1
1	1	1	-1	1	1
1	1	-1	1	1	1
1	-1	1	1	1	1
-1	1	1	1	1	1

Table 2: Example LaTeX Table

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	1	1	1	-1	-1
1	1	1	-1	1	-1
1	1	1	-1	-1	1
1	1	-1	1	1	-1
1	1	-1	1	-1	1
1	-1	1	1	1	-1
1	-1	1	-1	1	1
1	-1	-1	1	1	1
-1	1	1	1	-1	1
-1	1	1	-1	1	1
-1	1	-1	1	1	1
-1	-1	1	1	1	1

Table 3: Example LaTeX Table

Class 7
Class 8
Class 9
Class 10
Class 11

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	1	1	−1	−1	−1
1	−1	−1	1	1	−1
−1	1	−1	1	−1	1
−1	−1	1	−1	1	1

Table 4: Example LaTeX Table

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	1	−1	1	−1	−1
1	−1	1	−1	1	−1
−1	1	1	−1	−1	1
−1	−1	−1	1	1	1

Table 5: Example LaTeX Table

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	1	−1	−1	1	1
1	−1	1	1	−1	1
−1	1	1	1	1	−1

Table 6: Example LaTeX Table

What is the eigenspectrum of each symmetry class?

Do any of the symmetry classes have an eigenspectrum that is sign invariant? In other words, if you change the sign of each eigenvalue, does the overall set of eigenvalues remain unchanged?

As shown in Figure , the eigenspectrum of class 7 has sign flip invariance. It is because the Hamiltonian of class 7 is still in the class 7 even after sign flip operation.

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	1	−1	−1	1	−1
1	1	−1	−1	−1	1
1	−1	1	1	−1	−1
1	−1	1	−1	−1	1
1	−1	−1	1	−1	1
1	−1	−1	−1	1	1
−1	1	1	1	−1	−1
−1	1	1	−1	1	−1
−1	1	−1	1	1	−1
−1	1	−1	−1	1	1
−1	−1	1	1	1	−1
−1	−1	1	1	−1	1

Table 7: Example LaTeX Table

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	1	−1	−1	−1	−1
1	−1	1	−1	−1	−1
1	−1	−1	1	−1	−1
1	−1	−1	−1	1	−1
−1	1	1	−1	−1	−1
−1	1	−1	1	−1	−1
−1	1	−1	−1	−1	1
−1	−1	1	−1	1	−1
−1	−1	1	−1	−1	1
−1	−1	−1	1	1	−1
−1	−1	−1	1	−1	1
−1	−1	−1	−1	1	1

Table 8: Example LaTeX Table

Part 2 : Adiabatic evolution to the ground state

Plot the eigenspectra as a function of s for each symmetry class.

Class 1

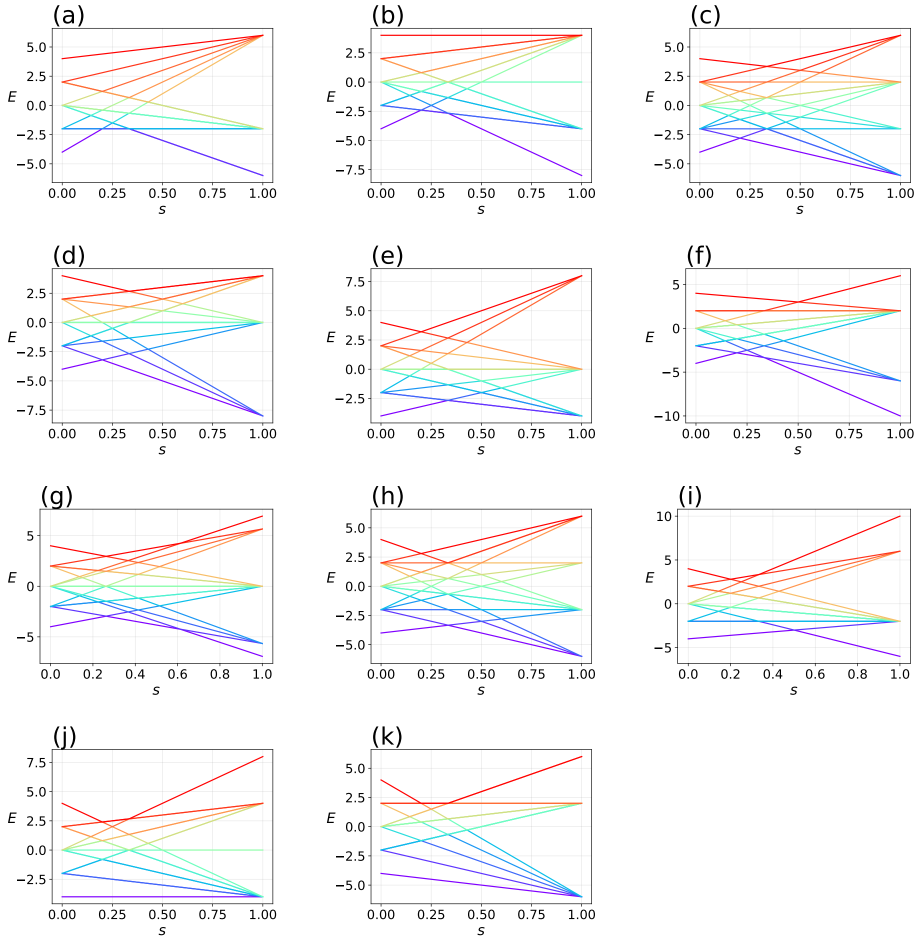

Figure 2: caption

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	−1	−1	−1	−1	1
−1	1	−1	−1	1	−1
−1	−1	1	1	−1	−1

Table 9: Example LaTeX Table

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
1	−1	−1	−1	−1	−1
−1	1	−1	−1	−1	−1
−1	−1	1	−1	−1	−1
−1	−1	−1	1	−1	−1
−1	−1	−1	−1	1	−1
−1	−1	−1	−1	−1	1

Table 10: Example LaTeX Table

J_{01}	J_{02}	J_{03}	J_{12}	J_{13}	J_{23}
−1	−1	−1	−1	−1	−1

Table 11: Example LaTeX Table

Plot the eigenspectrum for H_{sg} as a function of s.

Write a program to carry out adiabatic evolution, starting from the pure mixer Hamiltonian and ending with H_{sg}. Two parameters govern this program: the time τ over which the adiabatic evolution takes place and the number of time steps N.

Heisenberg Spin Glass

In order to match the target vector of sixteen probabilities by performing adiabatic evolution, what value do you need to use for τ and N? The exact target probabilities and the probabilities resulting from adiabatic evolution should match to three significant decimal digits.

Heisenberg Spin Glass

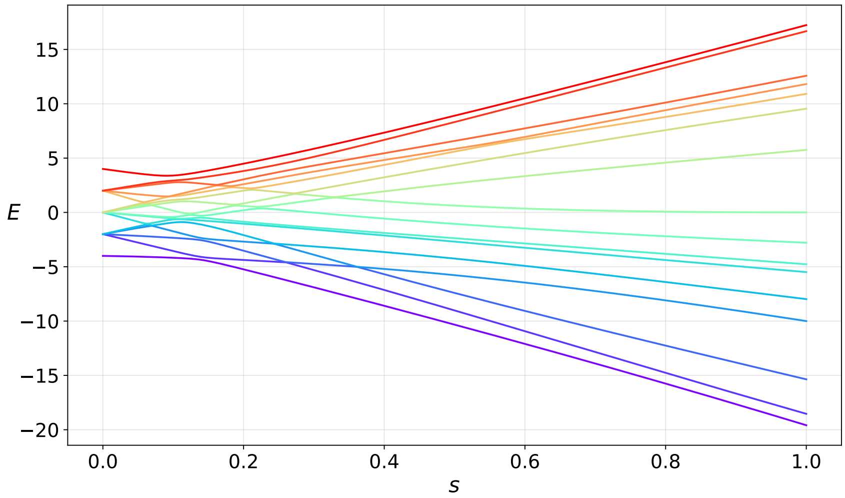

Figure 3

List out the terms in H_{sg} and H_m and, for each term, provide the gate that arises when that term is exponentiated. You can use any gate in Qiskit's standard gate set. How many single-qubit gates are needed for each time step? How many two-qubit gates are needed for each time step?

Heisenberg Spin Glass

Assess the circuit needed to carry out the adiabatic evolution. Roughly how many single-qubit gates and two-qubit gates would be required to achieve three-digit accuracy for the sixteen probabilities characterizing the ground state? Do you think this is practical?

Heisenberg Spin Glass

Part 3 : A simpler path to the ground state

Build a numerical program as outlined above, which contains arbitrary angle parameters and qubit pairs for XY-mixers. Experiment with the XY-mixers to obtain angle parameters and qubit pairs that rotate the final state so that its probabilities match the ground state probabilities of H_{sg}. This problem can be solved using three XY-mixers.

Heisenberg Spin Glass

Convert the numerical program into a quantum circuit. Verify that the output of this quantum circuit generates the correct probabilities for the ground state of H_{sg}.

Heisenberg Spin Glass

Check the phases of the non-zero components of the quantum circuit created in 3.b. Are they all real? If so, you can skip this step. Otherwise, find circuit elements that you can add to the end of your circuit so that all non-zero computational basis states have the same phase.

Heisenberg Spin Glass

Use your quantum circuit to measure the expectation value of the ground state of H_{sg} . You will have to repeat your circuit three times - once to measure in each of the X, Y and Z computational bases. Each term of H_{sg} can be measured in one of the three computational bases. Sum the contributions from the three bases and confirm that the summed energy is equal to the ground state eigenvalue

Heisenberg Spin Glass